

SHUBHAM PURI

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EDUCATION

New York Institute of Technology, Manhattan, NY,
B.S. in Computer Science

Anticipated May 2026

Relevant Courses: Data Structures, Design and Algorithms, Software Engineering, Artificial Intelligence, Operating Systems

SKILLS

Languages: Python, JavaScript, TypeScript, Java, SQL, PostgreSQL, React, React Native, Node.js, FastAPI, HTML/CSS

Tools & Platforms: Supabase, Firebase, Docker, Git/GitHub, REST APIs, CI/CD, Vercel, Figma, Jira, Google Suite

AI & Data: LangGraph, TensorFlow Lite, Machine Learning, Gemini API, Prompt Engineering, AI Quality Analysis

Methodologies: Agile, Object-Oriented Programming (OOP), Full Stack Development, Unit Testing

PROJECTS

[Autonomous Code Review and Testing Agent](#)

Python, LangGraph, tree-sitter, GitHub API, Docker | In Progress

- Built a LangGraph agentic system that autonomously generates unit test suites, executes them in isolated Docker sandboxes, self-corrects failures, and opens a GitHub pull request with **zero human input** after initialization.
- Implemented AST-based code understanding via tree-sitter across multiple languages, enabling context-aware test generation covering edge cases beyond simple prompt-based tools.
- Designed a self-correction feedback loop that diagnoses root causes of test failures, patches generated tests, and iterates until a configurable coverage threshold is met.

[CS Career Navigator - Personalized Career Path Tool](#)

Python, FastAPI, Supabase, REST API, Gemini API

- Researched career mentorship gap with 20+ CS students; deployed intelligent assessment tool to 50+ active users, achieving **58% 30-day retention**.
- Defined product roadmap and user stories across 4 Agile sprints; A/B tested recommendation changes, improving relevance by **40%** and reducing time-to-career-clarity to a single session.

[AI-Powered Malware Detection App](#)

React Native, TensorFlow Lite, Firebase, Python

- Built a full-stack mobile app detecting malware in Android APKs via on-device TFLite neural network, enabling **fully offline** scanning with no user data sent to external servers.
- Implemented a cloud-based REST API model update pipeline, ensuring the detection engine stays current without requiring app store updates, maintaining accuracy against emerging threats.
- Designed an AI agent workflow guiding users through scan results with contextual explanations and step-by-step remediation, reducing friction for non-technical audiences.

[Salon Booking Platform - Production Web App](#)

TypeScript, Node.js, Supabase, PostgreSQL, REST API, Vercel

- Built and deployed a **production full-stack** booking platform actively used by a real salon business — conducted user interviews with the owner to define core flows for appointments, service browsing, and admin management.
- Streamlined appointment booking from 7 steps to 3 through iterative client feedback cycles; applied OOP design patterns delivering a scalable, maintainable codebase aligned to real business operations.

WORK EXPERIENCE

[Queens Library - Tech Mentor](#)

Queens, NY | May 2025 - Present

- Manage 20 laptops and 15 iPads for the Teen Center, ensuring 100% device availability; designed and led a 6-week web development curriculum for 15+ teens using HTML, CSS, and JavaScript - **3 students** completed independent projects.
- Mentored teens through project ideation and debugging using Agile problem-solving approaches, fostering technical self-sufficiency and growth mindsets.

[Outlier - Freelance AI Quality Analyst](#)

Remote | Oct 2024 - Feb 2025

- Designed 25+ adversarial and edge-case prompts across code generation, reasoning, and factual accuracy tasks using Agile sprint-based evaluation cycles, contributing to a **20% improvement** in model accuracy.
- Surfaced systemic failure patterns and documented failure taxonomies, improving model robustness and accelerating the refinement cycle across multiple task categories.

[Mouse Squad - Tech Support](#)

Brooklyn, NY | Sep 2021 - May 2022

- Diagnosed hardware and software issues across a multi-device environment, reducing system downtime by **30%**; managed SQL-logged inventory for 50+ devices, ensuring uninterrupted staff access.

HACKATHONS & COMPETITIONS

[Nepali Hackathon](#)

March 2026 | Team

- Built a mental health web app for stigma-free early detection and peer support; competed among 83 teams in a 48-hour sprint.

[NYIT Hackathon](#)

Manhattan 2024 | Solo

- Built a real-time facial recognition system using Python and OpenCV; extended into a fully automated attendance pipeline with identity verification.